

SET 26

test> use college;

switched to db college

college> db.orderinfo.insertOne({

| cust_id: 123,

| cust_name: "abc",

| status: "A",

| price: 250

| })

{

acknowledged: true,

insertedId: ObjectId('6a005be64ef1f835aaabc114')

}

1] college> db.orderinfo.aggregate([

| { \$match: { status: "A" } },

| {

| \$group: {

| _id: "\$cust_name",

| average_price: { \$avg: "\$price" }

| }

| }

|])

[{ _id: 'abc', average_price: 250 }]

2] college> db.orderinfo.find(

| {

| price: { \$gte: 100, \$lte: 1000 }

| },

| {

| status: 1,

| _id: 0

| }

```
| )  
[ { status: 'A' } ]  
3] college> db.orderinfo.find(  
| {},  
| {  
|   _id: 0  
| }  
| )  
[ { cust_id: 123, cust_name: 'abc', status: 'A', price: 250 } ]  
4] college> db.orderinfo.createIndex({ cust_id: 1 })  
cust_id_1  
college> db.orderinfo.find({ cust_id: 123 })  
[  
  {  
    _id: ObjectId('6a005be64ef1f835aaabc114'),  
    cust_id: 123,  
    cust_name: 'abc',  
    status: 'A',  
    price: 250  
  }  
]
```

SET 27

```
college> db.movies.insertOne({
  |   name: "Movie1",
  |   type: "action",
  |   budget: 1000000,
  |   producer: {
  |     name: "producer1",
  |     address: "PUNE"
  |   }
  | })
{
  acknowledged: true,
  insertedId: ObjectId('6a005d3b4ef1f835aaabc118')
}
college> db.movies.find(
  | {
  |   budget: { $gt: 100000 }
  | },
  | {
  |   name: 1,
  |
  | }
  | )
[ { _id: ObjectId('6a005d3b4ef1f835aaabc118'), name: 'Movie1' } ]
1] college> db.movies.find(
  | {
  |   budget: { $gt: 100000 }
  | },
  | {
  |   name: 1,
  |   _id: 0
```

```

|   }
| )
[ { name: 'Movie1' } ]
2] college> db.movies.find(
|   {
|     "producer.address": "PUNE"
|   },
|   {
|     "producer.name": 1,
|     _id: 0
|   }
| )
[ { producer: { name: 'producer1' } } ]
3] college> db.movies.updateOne(
|   { type: "action" },
|   { $set: { type: "horror" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
or
college> db.movies.updateMany(
|   {
|     type: "action"
|   },
|   {
|     $set: { type: "horror" }

```

```

| }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
college> db.movies.find()
[
  {
    _id: ObjectId('6a005d3b4ef1f835aaabc118'),
    name: 'Movie1',
    type: 'horror',
    budget: 1000000,
    producer: { name: 'producer1', address: 'PUNE' }
  }
]

```

```

4] college> db.movies.find(
| {
|   "producer.name": "producer1"
| },
| {
|   name: 1,
|   "producer.address": 1,
|   _id: 0
| }
| )
[ { name: 'Movie1', producer: { address: 'PUNE' } } ]

```

SET 28

```
1] college> db.Teachers.insertMany([
| {
|   Tname: "Rahul",
|   dno: 1,
|   Experience: 5,
|   Salary: 12000,
|   Date_of_Joining: "2020-06-10"
| },
| {
|   Tname: "Sneha",
|   dno: 2,
|   Experience: 8,
|   Salary: 15000,
|   Date_of_Joining: "2018-04-15"
| },
| {
|   Tname: "Amit",
|   dno: 2,
|   Experience: 3,
|   Salary: 9000,
|   Date_of_Joining: "2022-01-12"
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0061614ef1f835aaabc119'),
    '1': ObjectId('6a0061614ef1f835aaabc11a'),
    '2': ObjectId('6a0061614ef1f835aaabc11b')
  }
}
```

```

}
college> db.Department.insertMany([
| {
|   Dno: 1,
|   Dname: "Computer"
| },
| {
|   Dno: 2,
|   Dname: "IT"
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a00616c4ef1f835aaabc11c'),
    '1': ObjectId('6a00616c4ef1f835aaabc11d')
  }
}
college> db.Students.insertMany([
| {
|   Sname: "xyz",
|   Roll_No: 2,
|   Class: "SE"
| },
| {
|   Sname: "Riya",
|   Roll_No: 5,
|   Class: "FE"
| },
| {
|   Sname: "Karan",

```

```

|   Roll_No: 7,
|   Class: "TE"
| }
| })
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a00618f4ef1f835aaabc11e'),
    '1': ObjectId('6a00618f4ef1f835aaabc11f'),
    '2': ObjectId('6a00618f4ef1f835aaabc120')
  }
}
2] college> db.Teachers.find(
| {
|   dno: 2,
|   Salary: { $gte: 10000 }
| }
| )
[
{
  _id: ObjectId('6a0061614ef1f835aaabc11a'),
  Tname: 'Sneha',
  dno: 2,
  Experience: 8,
  Salary: 15000,
  Date_of_Joining: '2018-04-15'
}
]
3] college> db.Students.find(
| {
|   $or: [

```

```

|   { Roll_No: 2 },
|   { Sname: "xyz" }
| ]
| }
| )
[
{
  _id: ObjectId('6a00618f4ef1f835aaabc11e'),
  Sname: 'xyz',
  Roll_No: 2,
  Class: 'SE'
}
]

```

```

4] college> db.Students.updateOne(

```

```

| {
|   Roll_No: 5
| },
| {
|   $set: { Sname: "Neha" }
| }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

college> db.Students.find()

```

```

[
{

```

```

    _id: ObjectId('6a00618f4ef1f835aaabc11e'),
    Sname: 'xyz',
    Roll_No: 2,
    Class: 'SE'
  },
  {
    _id: ObjectId('6a00618f4ef1f835aaabc11f'),
    Sname: 'Neha',
    Roll_No: 5,
    Class: 'FE'
  },
  {
    _id: ObjectId('6a00618f4ef1f835aaabc120'),
    Sname: 'Karan',
    Roll_No: 7,
    Class: 'TE'
  }
]
5] college> db.Students.deleteMany(
| {
|   Class: "FE"
| }
| )
{ acknowledged: true, deletedCount: 1 }
college> db.Students.find()
[
  {
    _id: ObjectId('6a00618f4ef1f835aaabc11e'),
    Sname: 'xyz',
    Roll_No: 2,
    Class: 'SE'
  }
]

```

```
},  
{  
  _id: ObjectId('6a00618f4ef1f835aaabc120'),  
  Sname: 'Karan',  
  Roll_No: 7,  
  Class: 'TE'  
}  
]  
6] college> db.Students.createIndex({ Roll_No: 1 })  
Roll_No_1  
college> db.Students.getIndexes()  
[  
  { v: 2, key: { _id: 1 }, name: '_id_' },  
  { v: 2, key: { Roll_No: 1 }, name: 'Roll_No_1' }  
]
```

SET 29

```
1] college> use department;
```

switched to db department

```
2] department> db.teacher.insertMany([
```

```
| {
```

```
|   name: "Rahul",
```

```
|   department: "Computer",
```

```
|   experience: 5,
```

```
|   salary: 25000
```

```
| },
```

```
| {
```

```
|   name: "Sneha",
```

```
|   department: "IT",
```

```
|   experience: 7,
```

```
|   salary: 30000
```

```
| },
```

```
| {
```

```
|   name: "Amit",
```

```
|   department: "Computer",
```

```
|   experience: 3,
```

```
|   salary: 20000
```

```
| ]])
```

```
{
```

```
  acknowledged: true,
```

```
  insertedIds: {
```

```
    '0': ObjectId('6a0063cf4ef1f835aaabc121'),
```

```
    '1': ObjectId('6a0063cf4ef1f835aaabc122'),
```

```
    '2': ObjectId('6a0063cf4ef1f835aaabc123')
```

```
  }
```

```
}
```

```
3] department> db.teacher.aggregate([
```

```
| {
|   $group:
|   {
|     _id: "$department",
|     average_salary: { $avg: "$salary" }
|   }
| }
| })
[
  { _id: 'Computer', average_salary: 22500 },
  { _id: 'IT', average_salary: 30000 }
]
```

4] department> db.teacher.aggregate([

```
| {
|   $group:
|   {
|     _id: "$department",
|     total_employees: { $sum: 1 }
|   }
| }
| })
[
  { _id: 'IT', total_employees: 1 },
  { _id: 'Computer', total_employees: 2 }
]
```

5] department> db.teacher.aggregate([

```
| {
|   $group:
|   {
|     _id: "$department",
|     minimum_salary: { $min: "$salary" }
|   }
| }
| })
```

```
| }  
| }  
| })  
[  
  { _id: 'IT', minimum_salary: 30000 },  
  { _id: 'Computer', minimum_salary: 20000 }  
]  
6] department> db.teacher.createIndex({ department: 1 })  
department_1  
department> db.teacher.getIndexes()  
[  
  { v: 2, key: { _id: 1 }, name: '_id_' },  
  { v: 2, key: { department: 1 }, name: 'department_1' }  
]  
department> db.teacher.dropIndex({ department: 1 })  
{ nIndexesWas: 2, ok: 1 }
```

SET 30

1] department> use product_detail

switched to db product_detail

2] product_detail> show dbs

admin 40.00 KiB

college 336.00 KiB

config 104.00 KiB

department 40.00 KiB

firstdb 8.00 KiB

local 40.00 KiB

3] product_detail> db.createCollection("product")

{ ok: 1 }

4] product_detail> show collections

product

5] product_detail> db.product.insertMany([

| {

| item: "pen",

| price: 10,

| quantity: 2

| },

| {

| item: "book",

| price: 50,

| quantity: 1

| },

| {

| item: "pen",

| price: 10,

| quantity: 5

| },

| {

```

|   item: "pencil",
|   price: 5,
|   quantity: 10
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0065f34ef1f835aaabc124'),
    '1': ObjectId('6a0065f34ef1f835aaabc125'),
    '2': ObjectId('6a0065f34ef1f835aaabc126'),
    '3': ObjectId('6a0065f34ef1f835aaabc127')
  }
}

```

```
6] product_detail> db.product.find()
```

```

[
  {
    _id: ObjectId('6a0065f34ef1f835aaabc124'),
    item: 'pen',
    price: 10,
    quantity: 2
  },
  {
    _id: ObjectId('6a0065f34ef1f835aaabc125'),
    item: 'book',
    price: 50,
    quantity: 1
  },
  {
    _id: ObjectId('6a0065f34ef1f835aaabc126'),
    item: 'pen',

```

```

    price: 10,
    quantity: 5
  },
  {
    _id: ObjectId('6a0065f34ef1f835aaaabc127'),
    item: 'pencil',
    price: 5,
    quantity: 10
  }
]
7] product_detail> db.product.createIndex({ item: 1 })
item_1
8] product_detail> db.product.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { item: 1 }, name: 'item_1' }
]
9] product_detail> db.product.createIndex(
| {
|   price: 1,
|   quantity: -1
| }
| )
price_1_quantity_-1
10] product_detail> db.product.dropIndex("price_1_quantity_-1")
{ nIndexesWas: 3, ok: 1 }

```

SET 31

```
college> db.orderinfo.insertOne({
  |   cust_id: 123,
  |   cust_name: "abc",
  |   status: "A",
  |   price: 250
  | })
{
  acknowledged: true,
  insertedId: ObjectId('6a007c0a948eaac10dabc115')
}
1] college> db.orderinfo.find(
  | {
  |   price: {
  |     $gte: 250,
  |     $lte: 450
  |   }
  | },
  | {
  |   cust_name: 1,
  |   _id: 0
  | }
  | )
[ { cust_name: 'abc' } ]
2] college> db.orderinfo.updateOne(
  | {
  |   cust_id: 123
  | },
  | {
  |   $inc: { price: 10 }
  | }
```

```

| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
college> db.orderinfo.find()
[
  {
    _id: ObjectId('6a007c0a948eaac10dabc115'),
    cust_id: 123,
    cust_name: 'abc',
    status: 'A',
    price: 260
  }
]
college> db.orderinfo.updateOne(
| {
|   cust_id: 124
| },
| {
|   $inc: { price: -5 }
| }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,

```

```

    upsertedCount: 0
  }
3] college> db.orderinfo.updateMany(
  | {},
  | {
  |   $unset: { status: "" }
  | }
  | )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
4] college> db.orderinfo.find()
[
  {
    _id: ObjectId('6a007c0a948eaac10dabc115'),
    cust_id: 123,
    cust_name: 'abc',
    price: 260
  }
]
5] college> db.orderinfo.find(
  | {
  |   $or: [
  |     { status: "A" },
  |     { price: 250 }
  |   ]
  | },

```

```
| {  
|   cust_name: 1,  
|   _id: 0  
| }  
| )
```

SET 33

```
1] test> use employee;
```

switched to db employee

```
2] employee> db.createCollection("emp1");
```

```
{ ok: 1 }
```

```
3] employee> db.emp1.insertMany([
```

```
| {
```

```
|   eno: 1,
```

```
|   ename: "Neha",
```

```
|   address: "Pune",
```

```
|   sal: 12000
```

```
| },
```

```
| {
```

```
|   eno: 2,
```

```
|   ename: "Rahul",
```

```
|   address: "Mumbai",
```

```
|   sal: 4000
```

```
| },
```

```
| {
```

```
|   eno: 3,
```

```
|   ename: "Nitin",
```

```
|   address: "Nashik",
```

```
|   sal: 18000
```

```
| },
```

```
| {
```

```
|   eno: 4,
```

```
|   ename: "Amit",
```

```
|   address: "Thane",
```

```
|   sal: 9000
```

```
| },
```

```
| {
```

```

|   eno: 5,
|   ename: "Nisha",
|   address: "Nagpur",
|   sal: 22000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a00b5b1764ac443c9abc114'),
    '1': ObjectId('6a00b5b1764ac443c9abc115'),
    '2': ObjectId('6a00b5b1764ac443c9abc116'),
    '3': ObjectId('6a00b5b1764ac443c9abc117'),
    '4': ObjectId('6a00b5b1764ac443c9abc118')
  }
}
4] employee> db.emp1.find()
[
  {
    _id: ObjectId('6a00b5b1764ac443c9abc114'),
    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 12000
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc115'),
    eno: 2,
    ename: 'Rahul',
    address: 'Mumbai',
    sal: 4000
  }
]

```

```

},
{
  _id: ObjectId('6a00b5b1764ac443c9abc116'),
  eno: 3,
  ename: 'Nitin',
  address: 'Nashik',
  sal: 18000
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc117'),
  eno: 4,
  ename: 'Amit',
  address: 'Thane',
  sal: 9000
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc118'),
  eno: 5,
  ename: 'Nisha',
  address: 'Nagpur',
  sal: 22000
}
]
5] employee> db.emp1.find(
| {
|   sal: { $gt: 5000 }
| }
| )
[
{
  _id: ObjectId('6a00b5b1764ac443c9abc114'),

```

```

    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 12000
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc116'),
    eno: 3,
    ename: 'Nitin',
    address: 'Nashik',
    sal: 18000
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc117'),
    eno: 4,
    ename: 'Amit',
    address: 'Thane',
    sal: 9000
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc118'),
    eno: 5,
    ename: 'Nisha',
    address: 'Nagpur',
    sal: 22000
  }
]
6] employee> db.emp1.find(
| {
|   sal: { $lt: 15000 }
| }

```

```

| )
[
{
  _id: ObjectId('6a00b5b1764ac443c9abc114'),
  eno: 1,
  ename: 'Neha',
  address: 'Pune',
  sal: 12000
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc115'),
  eno: 2,
  ename: 'Rahul',
  address: 'Mumbai',
  sal: 4000
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc117'),
  eno: 4,
  ename: 'Amit',
  address: 'Thane',
  sal: 9000
}
]
7] employee> db.emp1.find(
| {
|   sal: {
|     $gt: 10000,
|     $lt: 20000
|   }
| }

```

```
| )
[
{
  _id: ObjectId('6a00b5b1764ac443c9abc114'),
  eno: 1,
  ename: 'Neha',
  address: 'Pune',
  sal: 12000
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc116'),
  eno: 3,
  ename: 'Nitin',
  address: 'Nashik',
  sal: 18000
}
]
```

```
8] employee> db.emp1.updateMany(
```

```
| {},
| {
|   $mul: { sal: 1.10 }
| }
| )
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

```
employee> db.emp1.find()
```

```
[
  {
    _id: ObjectId('6a00b5b1764ac443c9abc114'),
    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 13200.0000000000002
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc115'),
    eno: 2,
    ename: 'Rahul',
    address: 'Mumbai',
    sal: 4400
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc116'),
    eno: 3,
    ename: 'Nitin',
    address: 'Nashik',
    sal: 19800
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc117'),
    eno: 4,
    ename: 'Amit',
    address: 'Thane',
    sal: 9900
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc118'),
```

```

    eno: 5,
    ename: 'Nisha',
    address: 'Nagpur',
    sal: 24200.0000000000004
  }
]
9] employee> db.emp1.deleteMany(
| {
|   sal: { $lt: 5000 }
| }
| )
{ acknowledged: true, deletedCount: 1 }
employee> db.emp1.find()
[
  {
    _id: ObjectId('6a00b5b1764ac443c9abc114'),
    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 13200.0000000000002
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc116'),
    eno: 3,
    ename: 'Nitin',
    address: 'Nashik',
    sal: 19800
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc117'),
    eno: 4,

```

```

    ename: 'Amit',
    address: 'Thane',
    sal: 9900
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc118'),
    eno: 5,
    ename: 'Nisha',
    address: 'Nagpur',
    sal: 24200.00000000000004
  }
]
10] employee> db.emp1.renameCollection("employee1")
{ ok: 1 }
11] employee> db.employee1.find(
| {
|   ename: /^n/i
| }
| )
[
  {
    _id: ObjectId('6a00b5b1764ac443c9abc114'),
    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 13200.00000000000002
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc116'),
    eno: 3,
    ename: 'Nitin',

```

```
    address: 'Nashik',
    sal: 19800
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc118'),
    eno: 5,
    ename: 'Nisha',
    address: 'Nagpur',
    sal: 24200.000000000004
  }
]
```

12] employee> db.employee1.find().sort(

```
| {
|   ename: 1
| }
| )
[
  {
    _id: ObjectId('6a00b5b1764ac443c9abc117'),
    eno: 4,
    ename: 'Amit',
    address: 'Thane',
    sal: 9900
  },
  {
    _id: ObjectId('6a00b5b1764ac443c9abc114'),
    eno: 1,
    ename: 'Neha',
    address: 'Pune',
    sal: 13200.000000000002
  },
]
```

```

{
  _id: ObjectId('6a00b5b1764ac443c9abc118'),
  eno: 5,
  ename: 'Nisha',
  address: 'Nagpur',
  sal: 24200.00000000000004
},
{
  _id: ObjectId('6a00b5b1764ac443c9abc116'),
  eno: 3,
  ename: 'Nitin',
  address: 'Nashik',
  sal: 19800
}
]

13] employee> use Employee1
switched to db Employee1

14] Employee1> db.dropDatabase()
{ ok: 1, dropped: 'Employee1' }

15] Employee1> db.emp1.insertOne(
| {
|   eno: 101,
|   ename: "Riya",
|   address: "Pune",
|   sal: 15000
| }
| )
{
  acknowledged: true,
  insertedId: ObjectId('6a00b690764ac443c9abc119')
}

```

```
Employee1> db.emp1.find()
```

```
[  
  {  
    _id: ObjectId('6a00b690764ac443c9abc119'),  
    eno: 101,  
    ename: 'Riya',  
    address: 'Pune',  
    sal: 15000  
  }  
]
```

```
16] Employee1> db.emp1.drop()
```

```
true
```